Do Local Chiefs Grant More Discretionary Tax Exemptions? Collector Identity and Property-Tax Exemptions in Kananga, DRC

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1 OLS

$$\text{Exempt}_{ij} = \beta_0 + \beta_1 \text{Local}_j + X_i'\gamma + (\text{Local}_j \times X_i)'\delta + \alpha_{s(j)} + \mu_{h(i)} + \tau_{t(i)} + \varepsilon_{ij}$$

where i is a property, j its neighbourhood, $\text{Local}_j = 1$ if chiefs collect (tmt= 2) and 0 if state agents collect (tmt= 1); X_i = baseline age, female, education, log income, HH size, born in Kananga, works for government, renters, assets, Luluwa, co-ethnic with collector (centred); α_s stratum, μ_h house-type and τ_t two-month time fixed effects. SEs clustered by neighbourhood.

Table 1: OLS: effect of Local (vs Central) collection on tax exemptions

	(1) Exempt	(2) Exempt	(3) Exempt	(4) Senior	(5) Widow	(6) Unjust. senior
Local	0.002	0.056	0.045	0.075^{**}	-0.027	0.030
	(0.045)	(0.042)	(0.043)	(0.037)	(0.025)	(0.020)
Age		0.012^{***}	0.011^{***}	0.009^{***}	0.002	-0.002^{***}
		(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
Co-ethnic		-0.127^{***}	-0.120^{**}	-0.086^{*}	-0.020	-0.016
		(0.043)	(0.057)	(0.051)	(0.037)	(0.029)
Female		0.107^{***}	0.126^{***}	-0.046	0.200^{***}	0.014
		(0.034)	(0.048)	(0.043)	(0.046)	(0.024)
Local × Age			0.001	0.001	-0.001	0.000
			(0.002)	(0.002)	(0.001)	(0.001)
Local × Co-ethnic			0.039	-0.095	0.080	-0.046
			(0.095)	(0.084)	(0.059)	(0.043)
Local × Female			-0.040	0.009	-0.052	0.003
			(0.067)	(0.058)	(0.057)	(0.035)
Local × Log income			-0.012	-0.006	-0.013^{*}	-0.004
			(0.012)	(0.010)	(0.008)	(0.004)
Local × Assets			0.013	-0.007	0.026	0.009
			(0.022)	(0.023)	(0.018)	(0.013)
Baseline covariates	No	Yes	Yes	Yes	Yes	Yes
Local × covariates	No	No	Yes	Yes	Yes	Yes
Stratum, house, time FE	Yes	Yes	Yes	Yes	Yes	Yes
Central mean	0.384	0.384	0.384	0.234	0.124	0.043
Observations	1,163	1,163	1,163	1,163	1,163	1,160
Clusters	172	172	172	172	172	172
R^2	0.089	0.258	0.266	0.230	0.161	0.128

Notes: Unit is a property registered by a collector in a Central or Local neighbourhood and matched to the 2017 baseline survey. Exempt = collector recorded the property as tax-exempt; Senior/Widow = exempted for that reason; Unjust. senior = exempted as a senior although the owner was under 55 at baseline. Covariates are centred at the sample mean, so Local is the effect for an average household. Columns (2)–(6) also control for HH size, education, log income, born in Kananga, works for government, renters, assets, Luluwa and missing-value dummies; columns (3)–(6) interact every covariate with Local. Only key coefficients are shown. Standard errors clustered by neighbourhood in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

2 Ridge

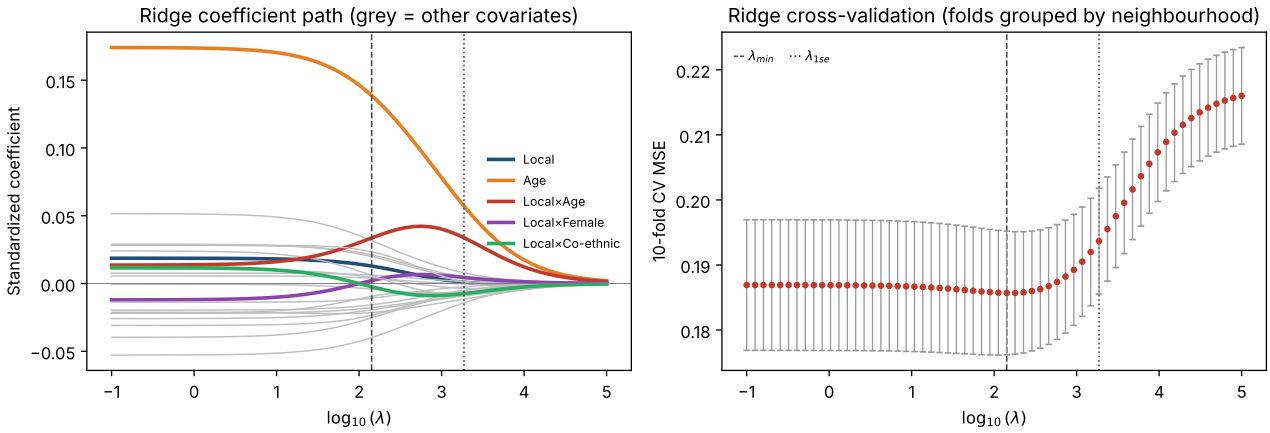


Figure 1: Ridge: coefficient path (left) and 10-fold CV MSE with ± 1 SE bars (right). Outcome: Exempt; 23 standardized features (Local, 11 covariates, 11 Local$\times$covariate interactions); fixed effects and missing dummies partialled out. Dashed: λ_{min}; dotted: λ_{1se}.

3 LASSO

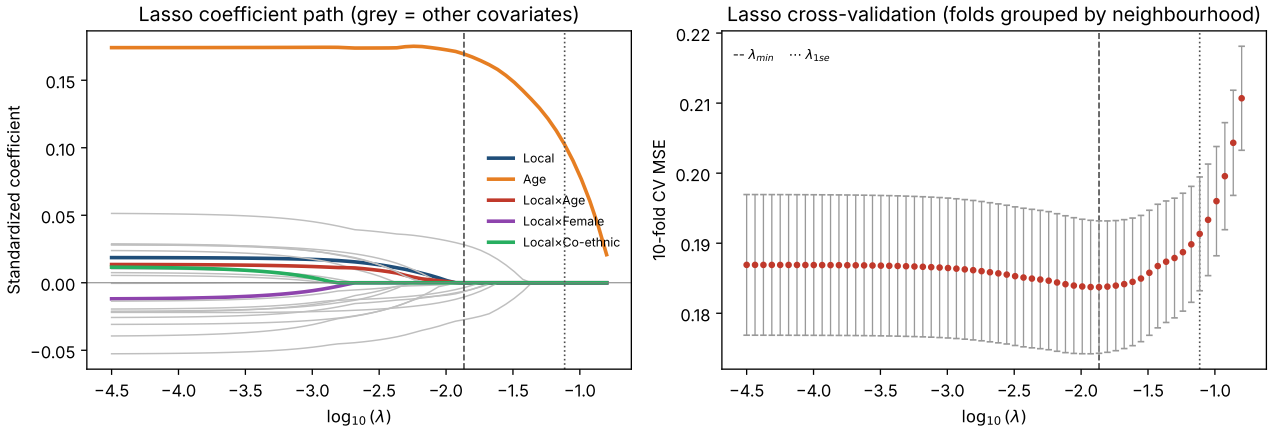


Figure 2: Lasso: coefficient path (left) and 10-fold CV MSE with ± 1 SE bars (right). Same features, sample and folds as Figure 1.

Findings. All three models agree that *age* is by far the strongest predictor of being exempted, while the Local treatment and its interactions are small: in OLS, Local raises exemptions for seniors by 7.5 pp ($p < 0.05$) but not for widows, and no Local$\times$covariate term, including Local$\times$co-ethnic, is significant. Lasso sharpens this: at λ_{min} it sets Local, Local$\times$Age and Local$\times$Co-ethnic to exactly zero (keeping only age, female, co-ethnic, assets and three tiny interactions), and at λ_{1se} only age survives, so the heterogeneity terms add almost no predictive power. This tells us more than OLS alone: any favouritism by chiefs is too small or too non-linear to show up in linear interactions, which motivates the next step of estimating heterogeneity flexibly with a causal forest.